

Proteins with multiple G protein-coupled receptor domains

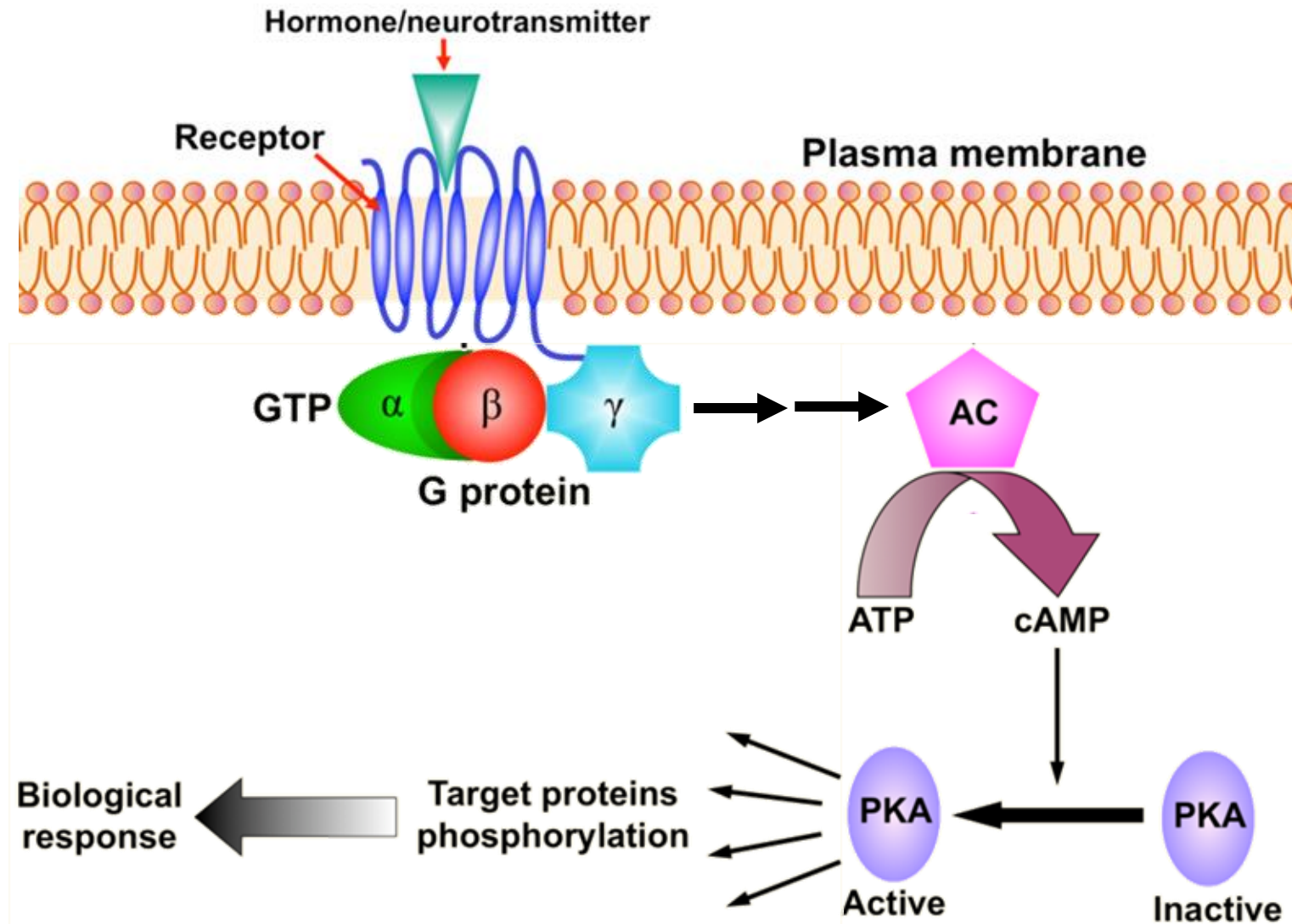
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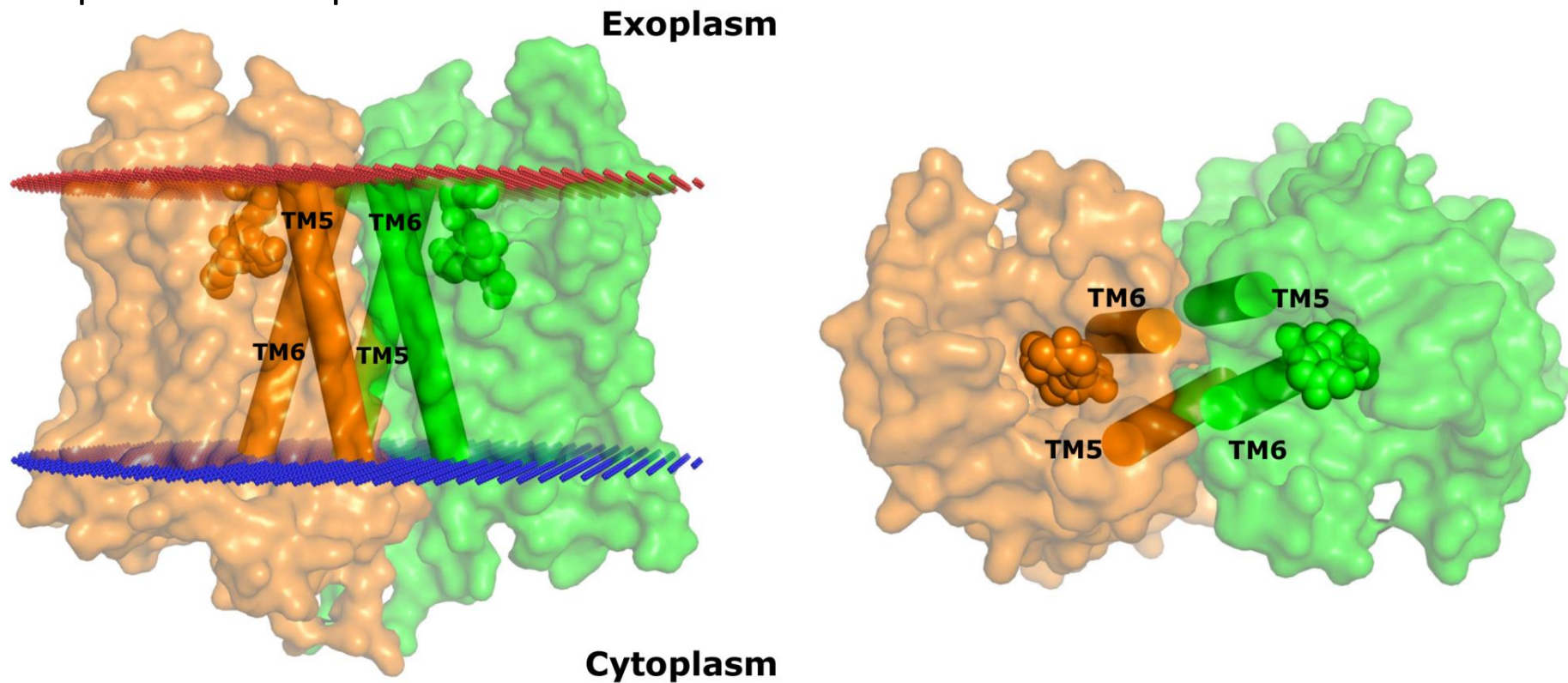
GPCR signaling



Many GPCRs form homo- or hetero-dimers

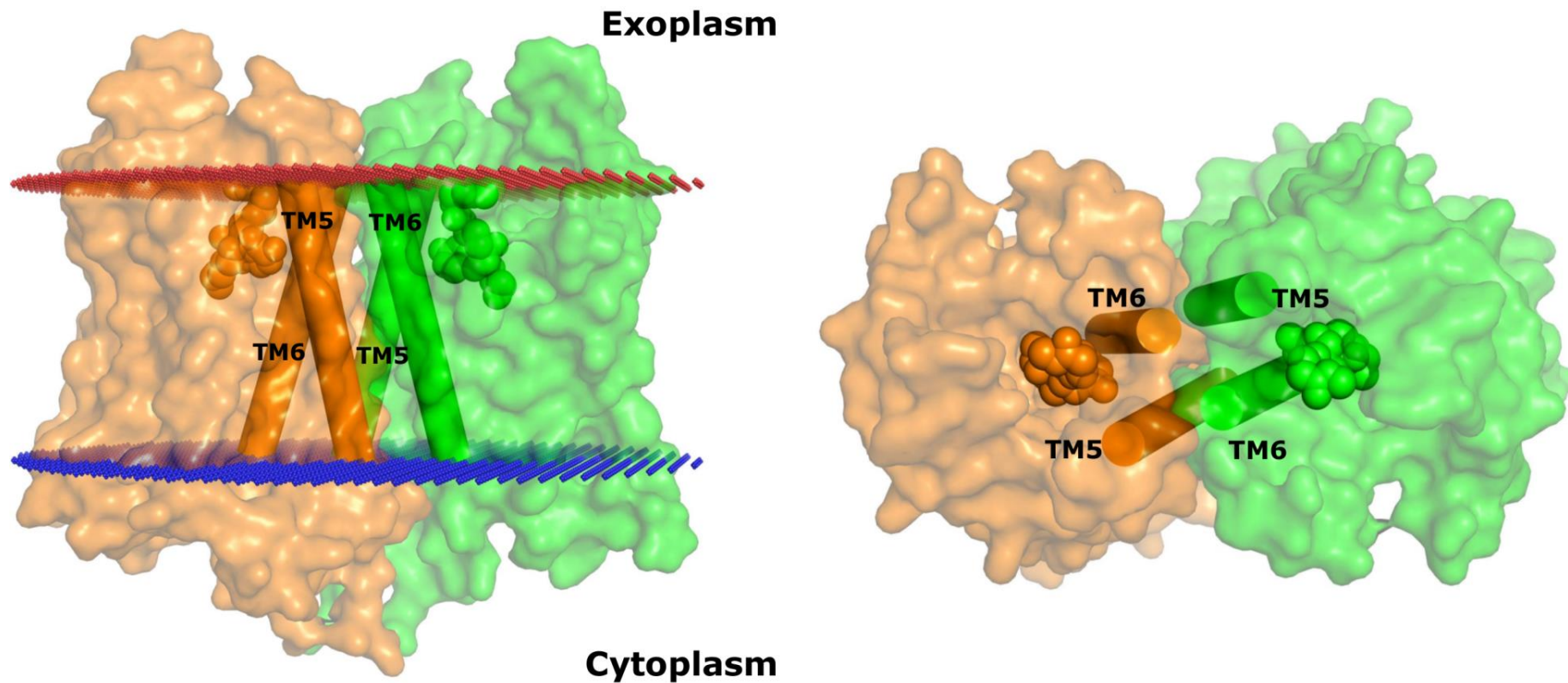
- Class A – mostly **regulation** (cooperativity, crosstalk)
- Class C – each chain may have **different roles**

μ -opioid receptor:



Many GPCRs form homo- or hetero-dimers

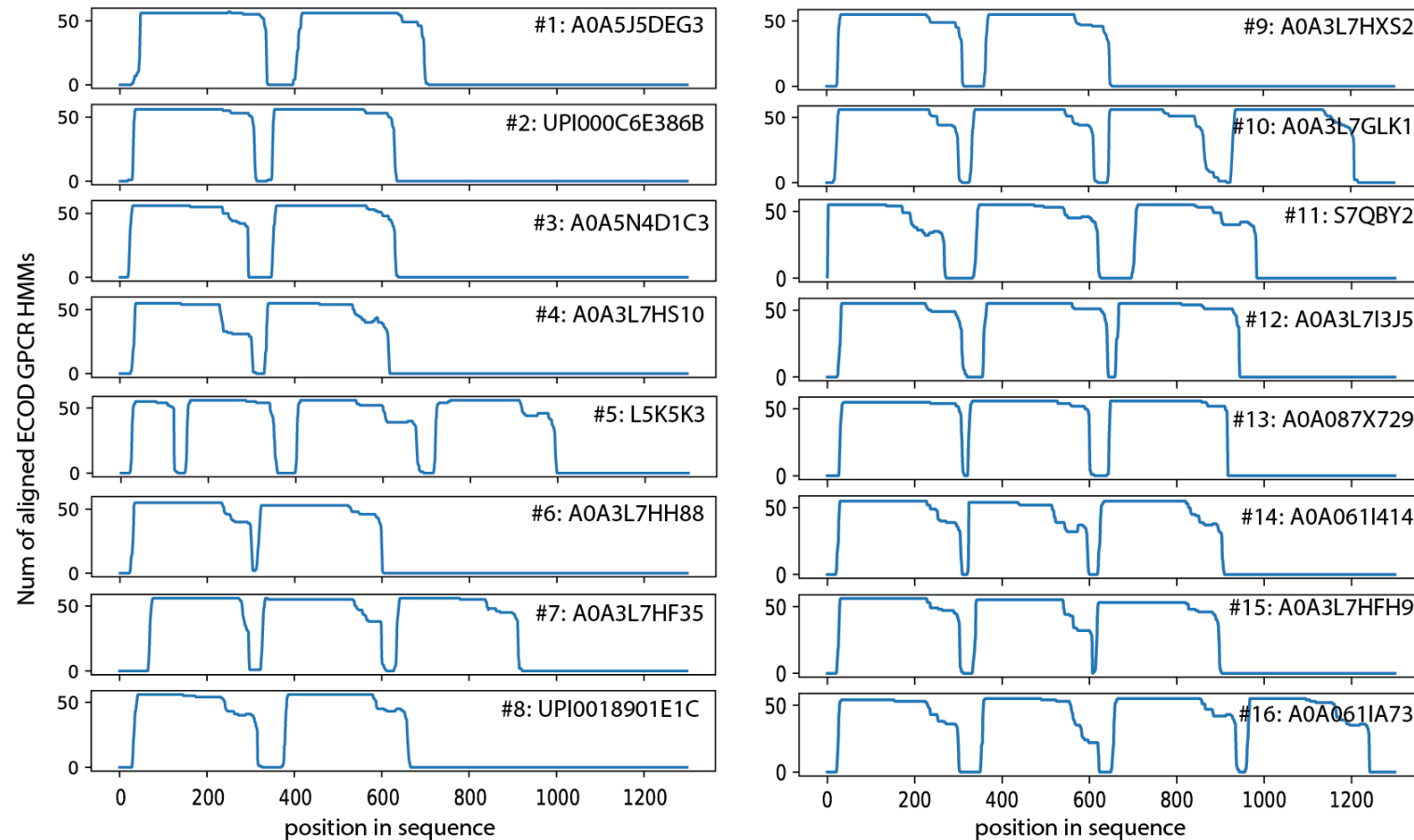
Are there receptors with two-GPCR domains within the same protein chain?



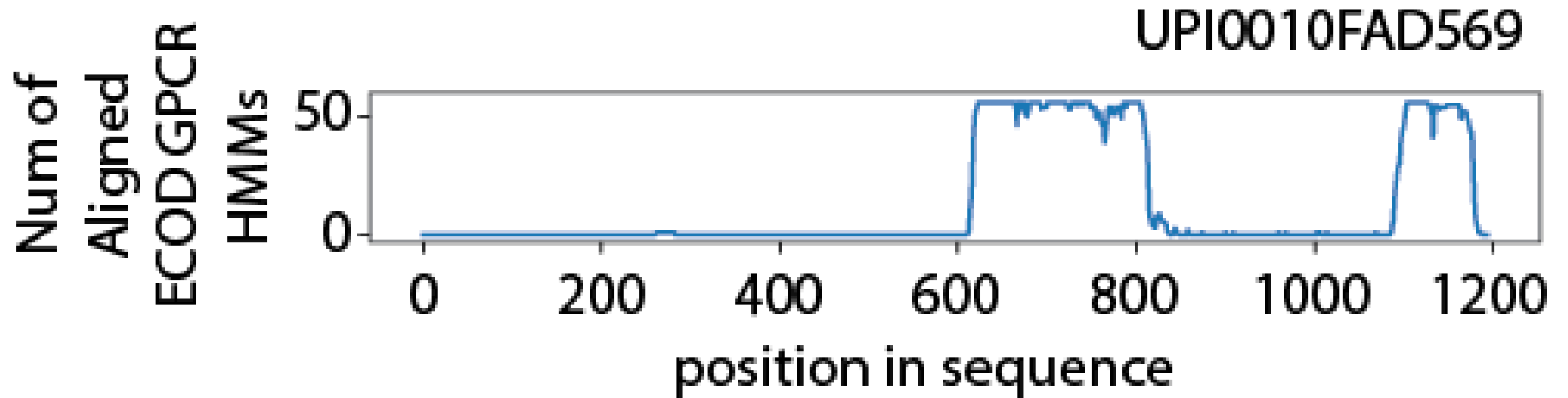
Finding proteins with multi-GPCR domains

- Constructed HMMER profiles for all 92 GPCR domains in the (99% NR) ECOD database
- Searched UniRef90 for proteins with non-overlapping multiple GPCR domains
- Clustered the hits using CD-HIT: found 1395 clusters
- Selected the 16 clusters with >5 hits and identical number of domains: 57 proteins in total
- Mapped them to representative genomes

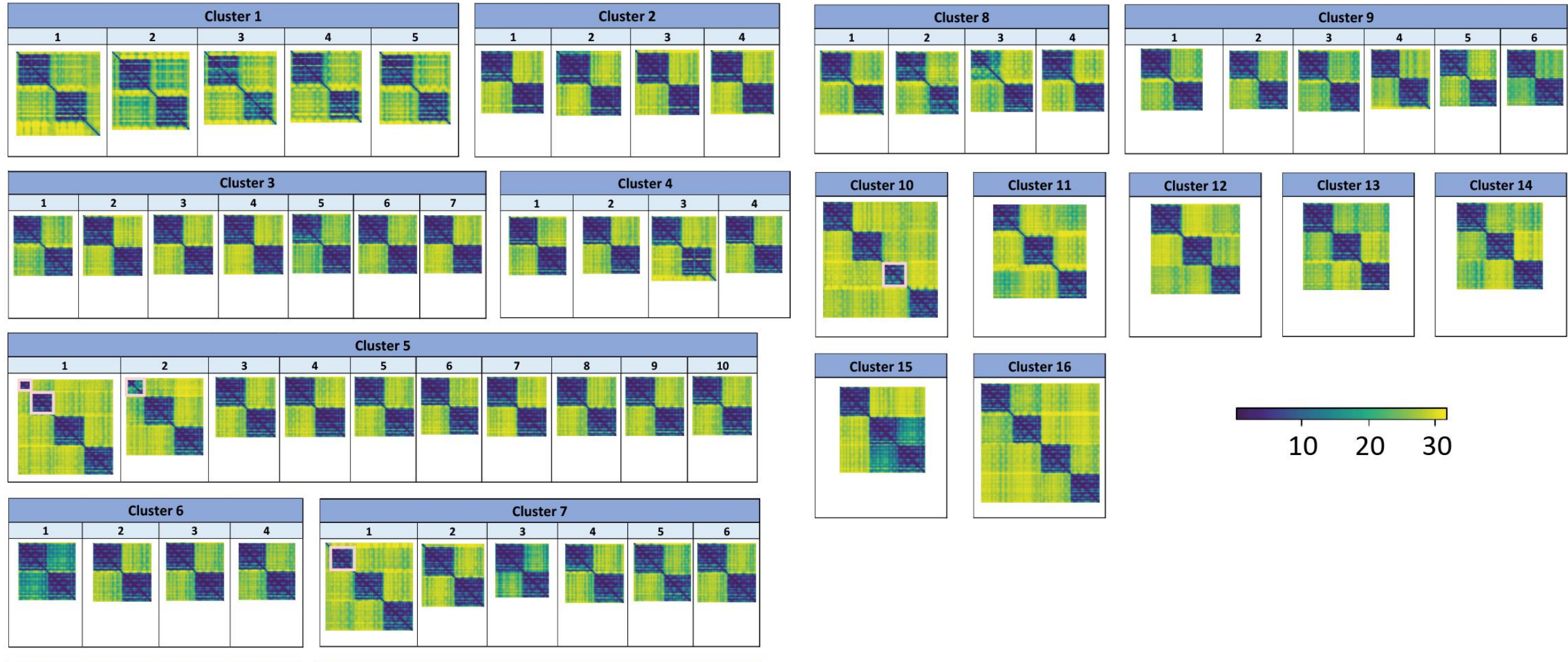
The number of ECOD GPCR HMMs aligned to each position in the sixteen clusters



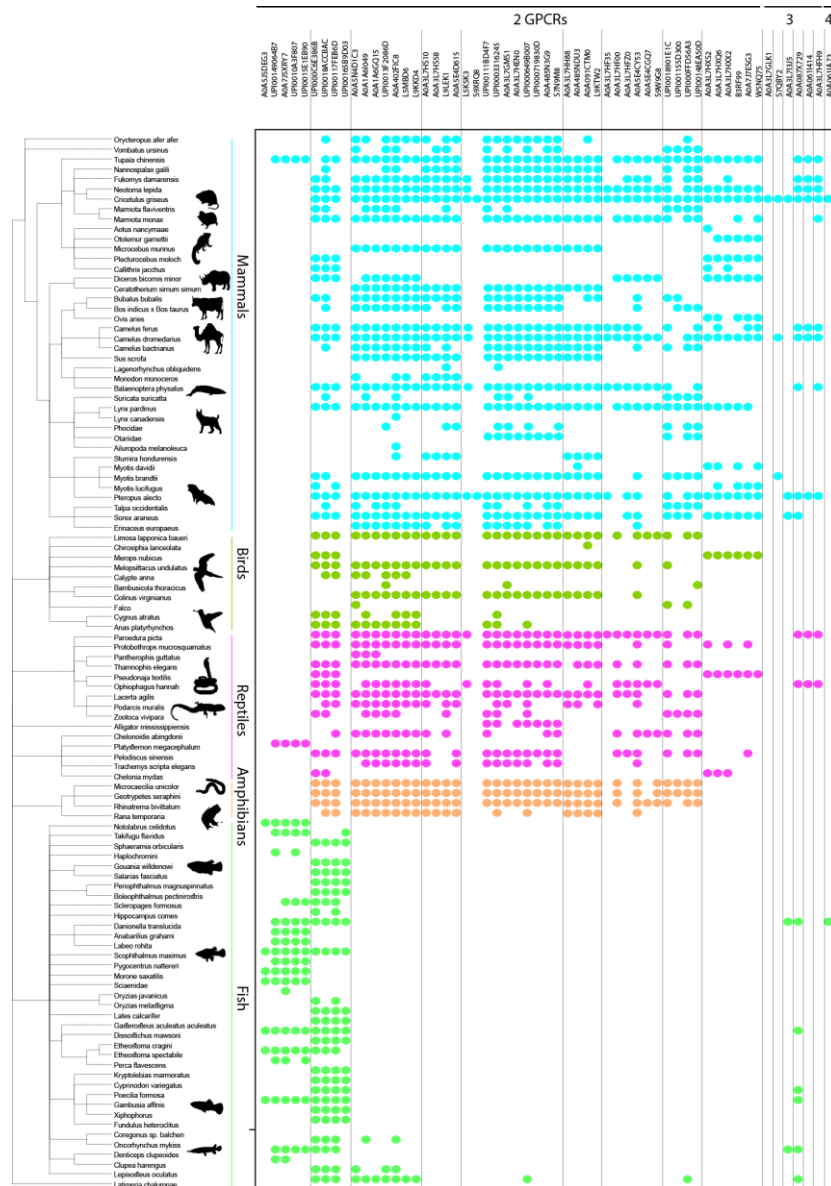
UPI0010FAD569 was filtered out because it has only one full GPCR domain



AlphaFold predicted pairwise aligned errors for the 57 multi-GPCR proteins



Tree of life distribution of homologues of the 57 multi-GPCR proteins



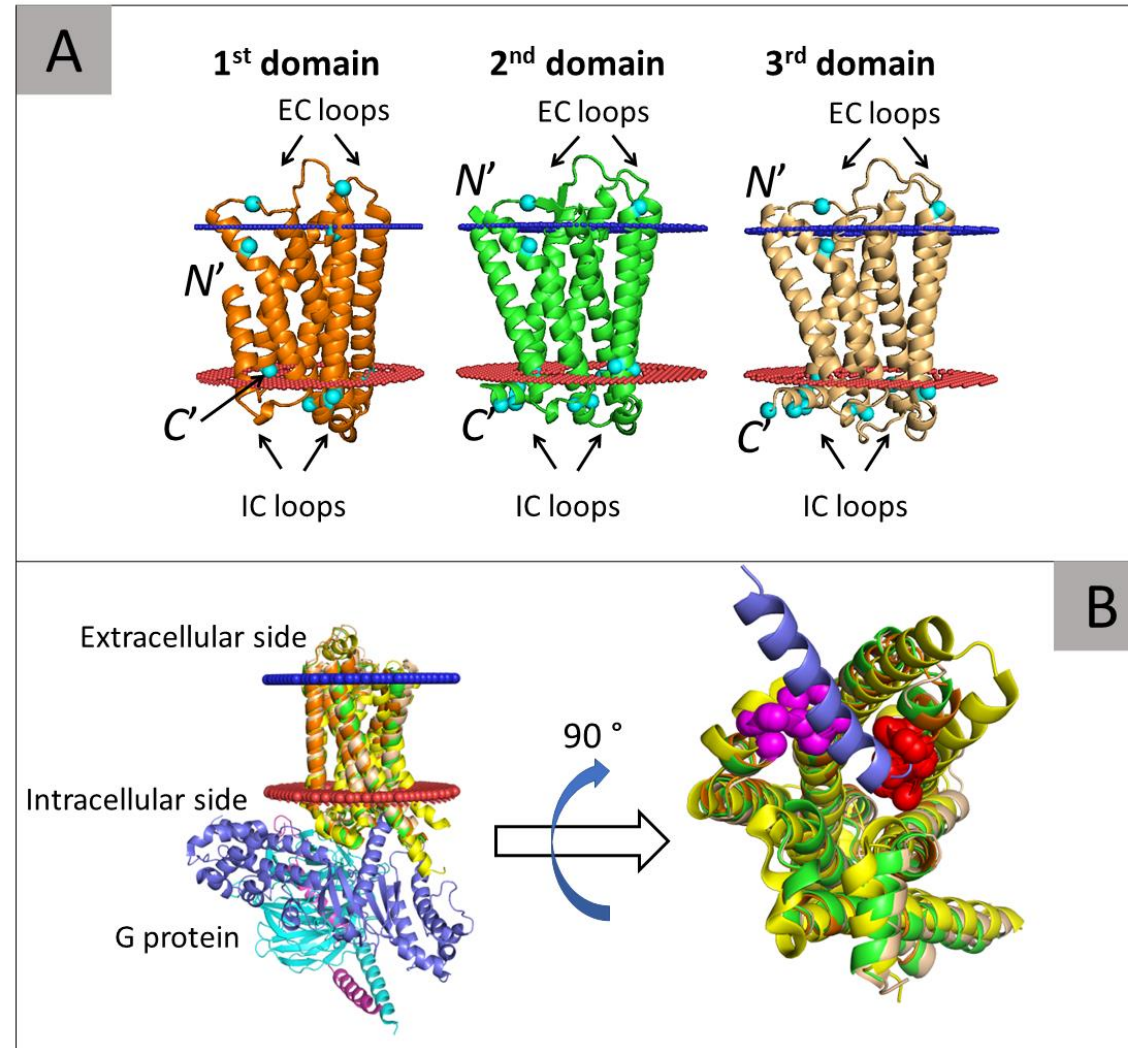
Single-domain GPCRs are broadly distributed in eukaryotes: primitive yeast and choanoflagellate → animals, **but multi-GPCRs only in Chordata species**

Multi-GPCR domains are only homologous to muscarinic acetylcholine receptor M3, adenosine receptor, rhodopsin, and orexin receptor type 2/ chemokine receptor type 4.

By homology, most are olfactory receptors, and some are chemokine receptors

Isildayancan, Kessel, Solan, Kolody, Ben-Tal. *BioRxiv*.

AlphaFold model of tri-GPCR-domain S7QBY2



The β_2 -AR receptor (yellow) in complex with G protein. α (blue), β (cyan), and γ (magenta)

The VAICxPLxY (pink) and KAFSTCASH (red) motifs

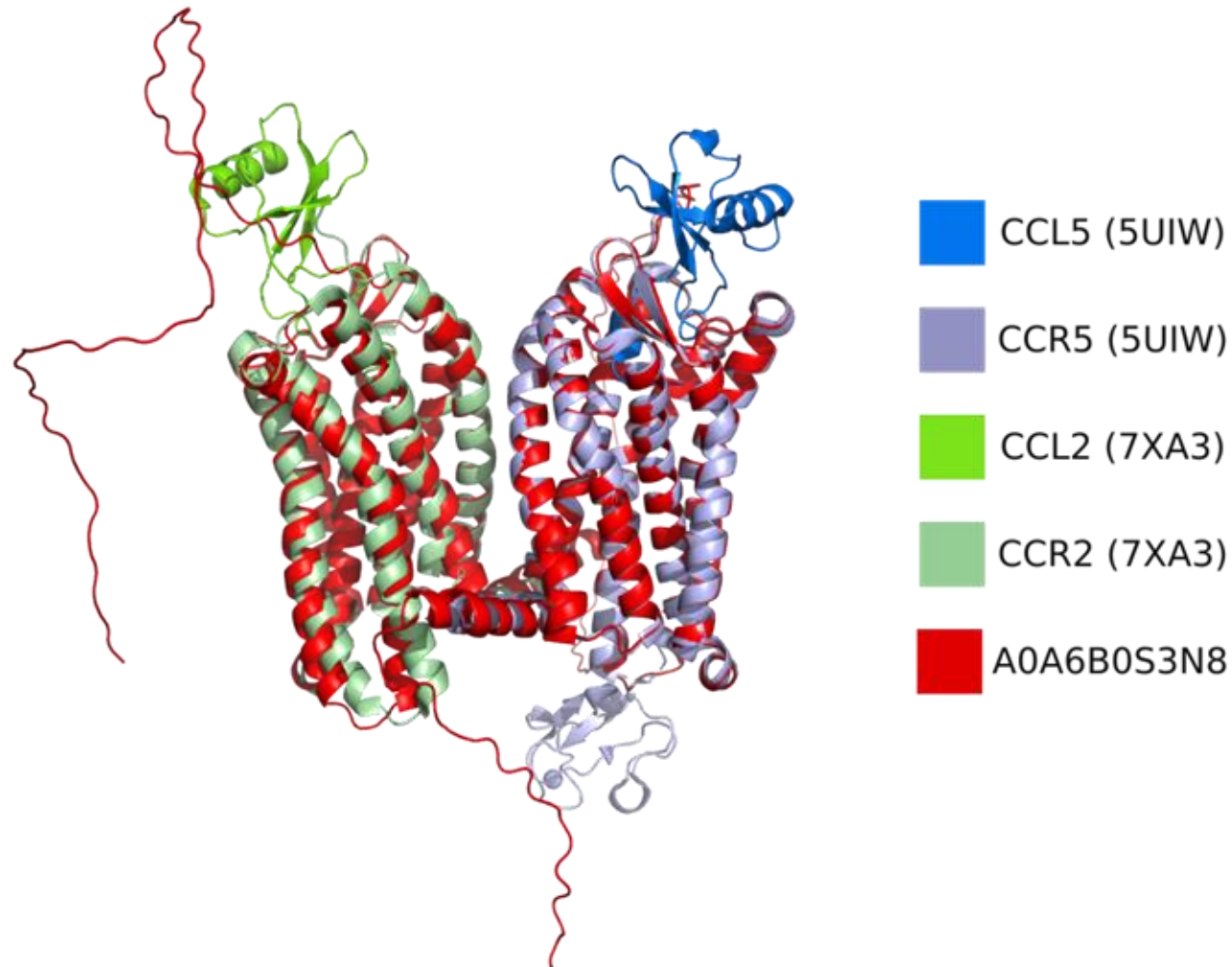
The A0A6B0S3N8 bi-GPCR from *Bos mutus* (Wild Yak): A chemokine receptor?



N-term domain is closest to **C-C Chemokine Receptor 2**: 84% seq identity, 100% coverage

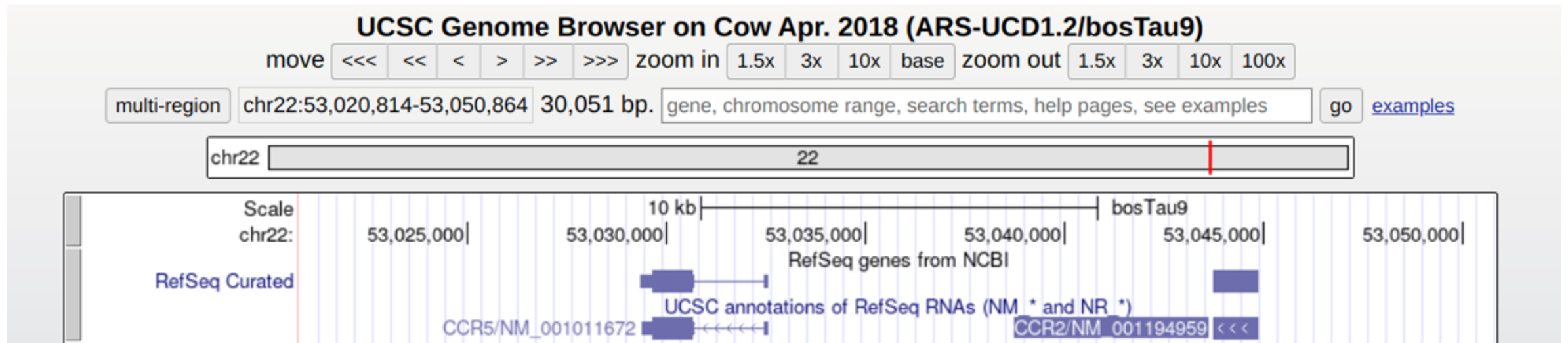
C-term domain is closest to **C-C Chemokine Receptor 5**: 99.6% seq identity, 100% coverage

Tentative complex of A0A6B0S3N8 with chemokine ligands



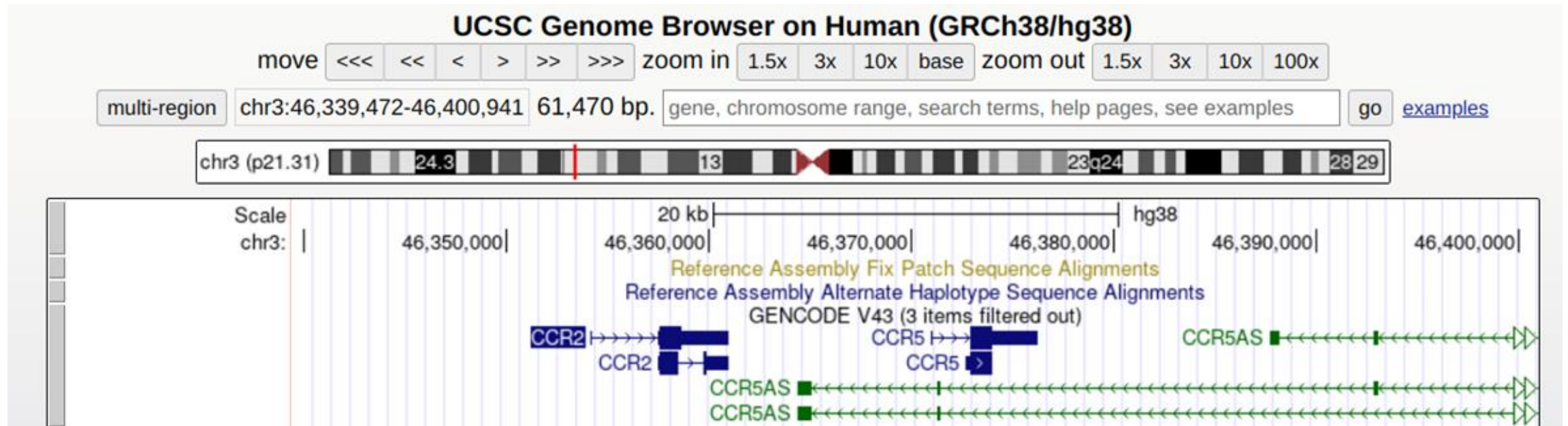
Superimposition of the AlphaFold model of A0A6B0S3N8 and two PDB structures: CCR2-CCL2 (7XA3; RMSD=1.4 Å) and CCR5-CCL5 (5UIW, RMSD=0.5 Å) complexes

The two chemokine receptors are near one another in the Cow genome



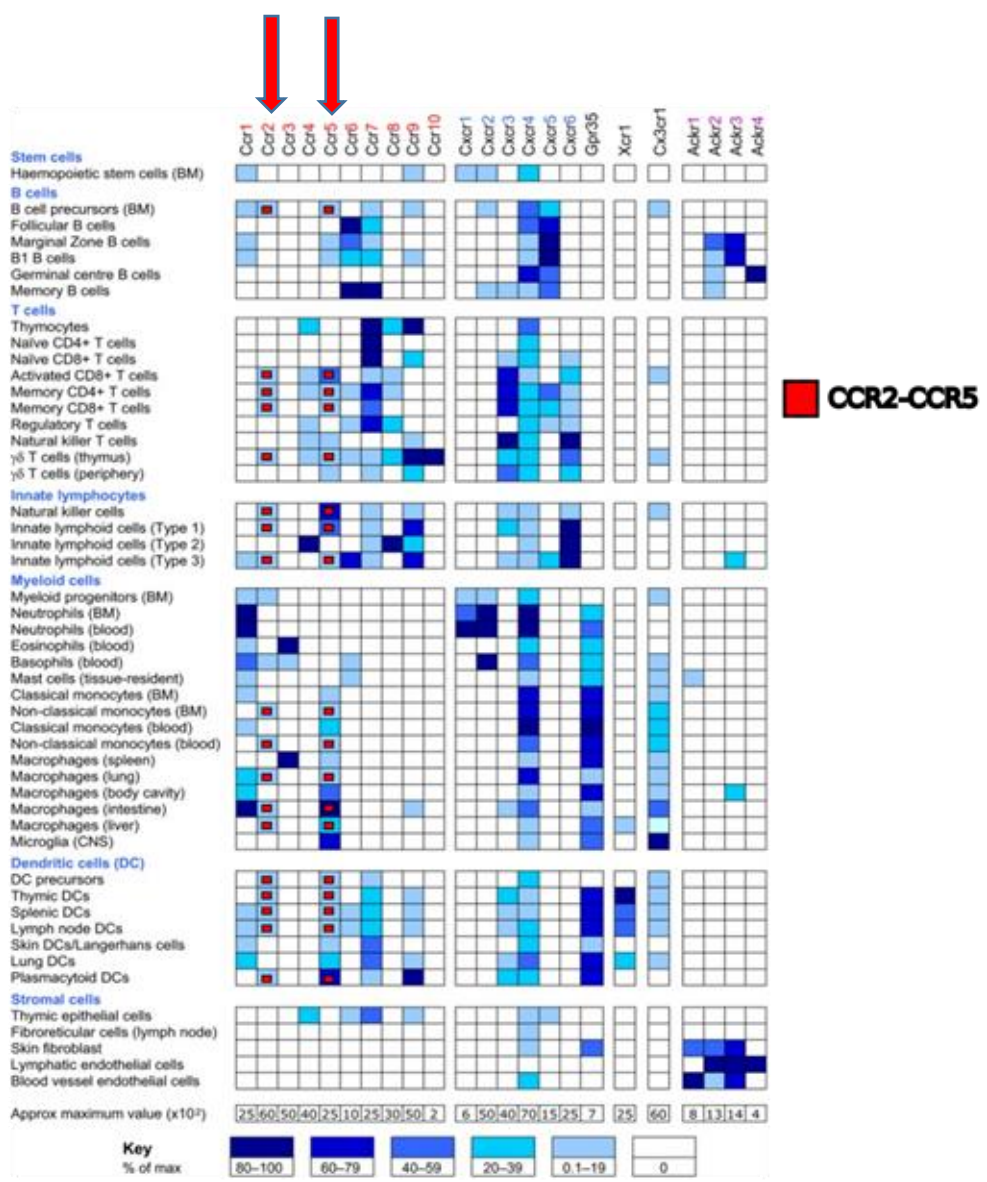
~10k bp
distance

The two chemokine receptors are near one another in the Human genome



~10k bp
distance

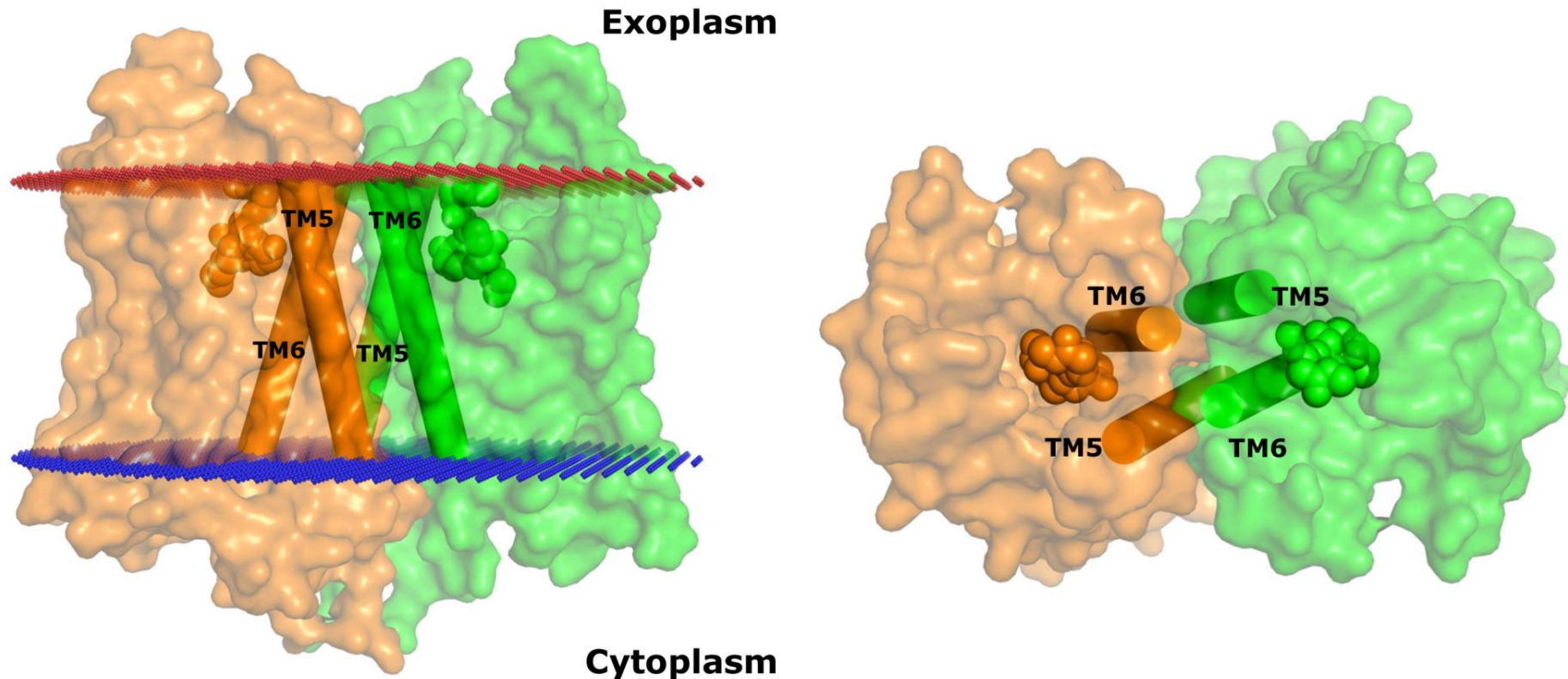
CCR2 and CCR5 co-express in the human immune system



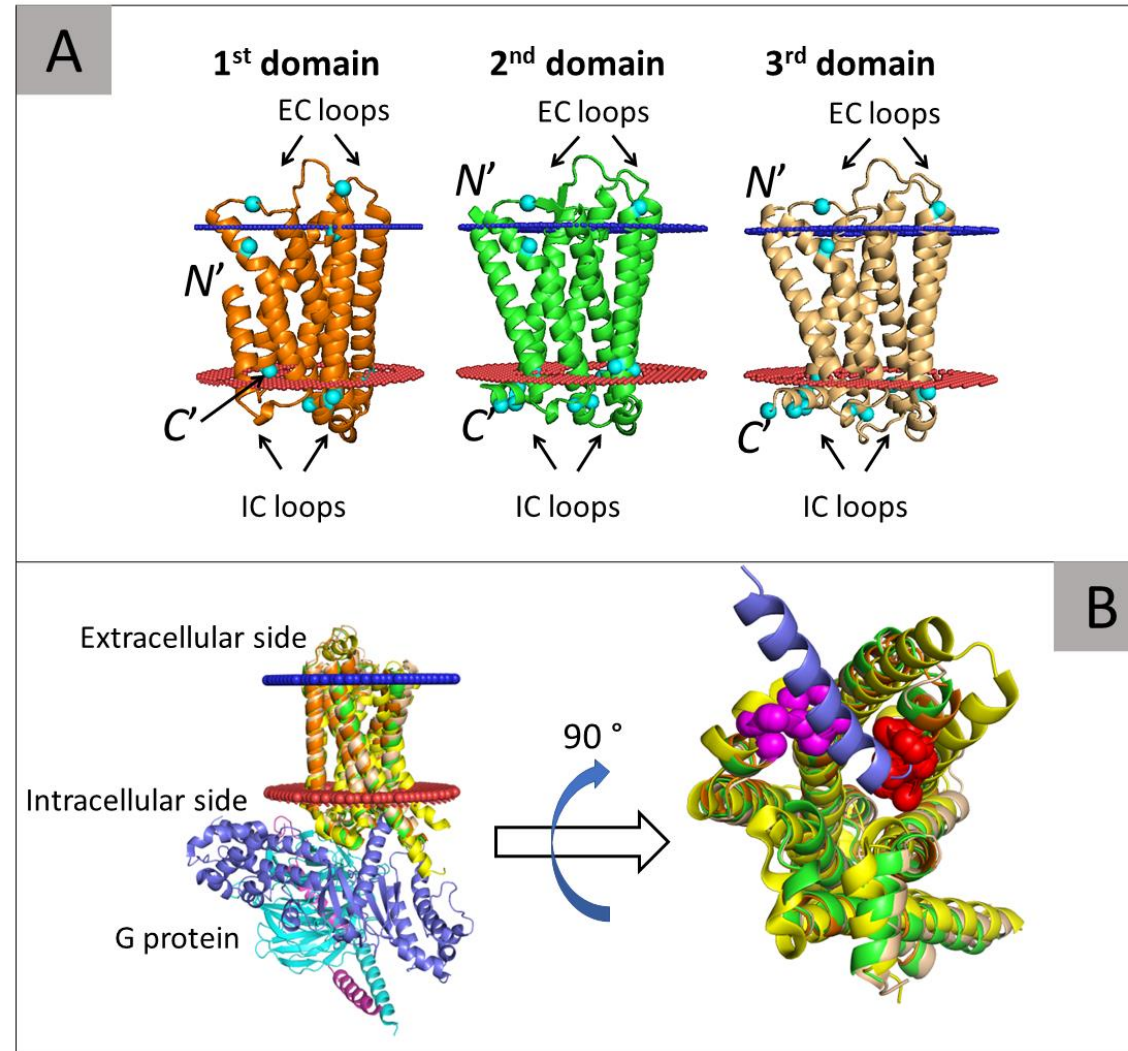
Adapted from Hughes CE, Nibbs RJB. A guide to chemokines and their receptors. FEBS J. 2018 Aug;285(16):2944-2971.

But proper GPCR function requires an N-terminal out topology...

As in homo- or hetero-dimers



Membrane topology and the positive-inside rule



● K, R

The β_2 -AR receptor (yellow) in complex with G protein. α (blue), β (cyan), and γ (magenta)

The VAICxPLxY (pink) and KAFSTCASH (red) motifs

Unsatisfied backbone hydrogen bonds in the membrane are energetically costly

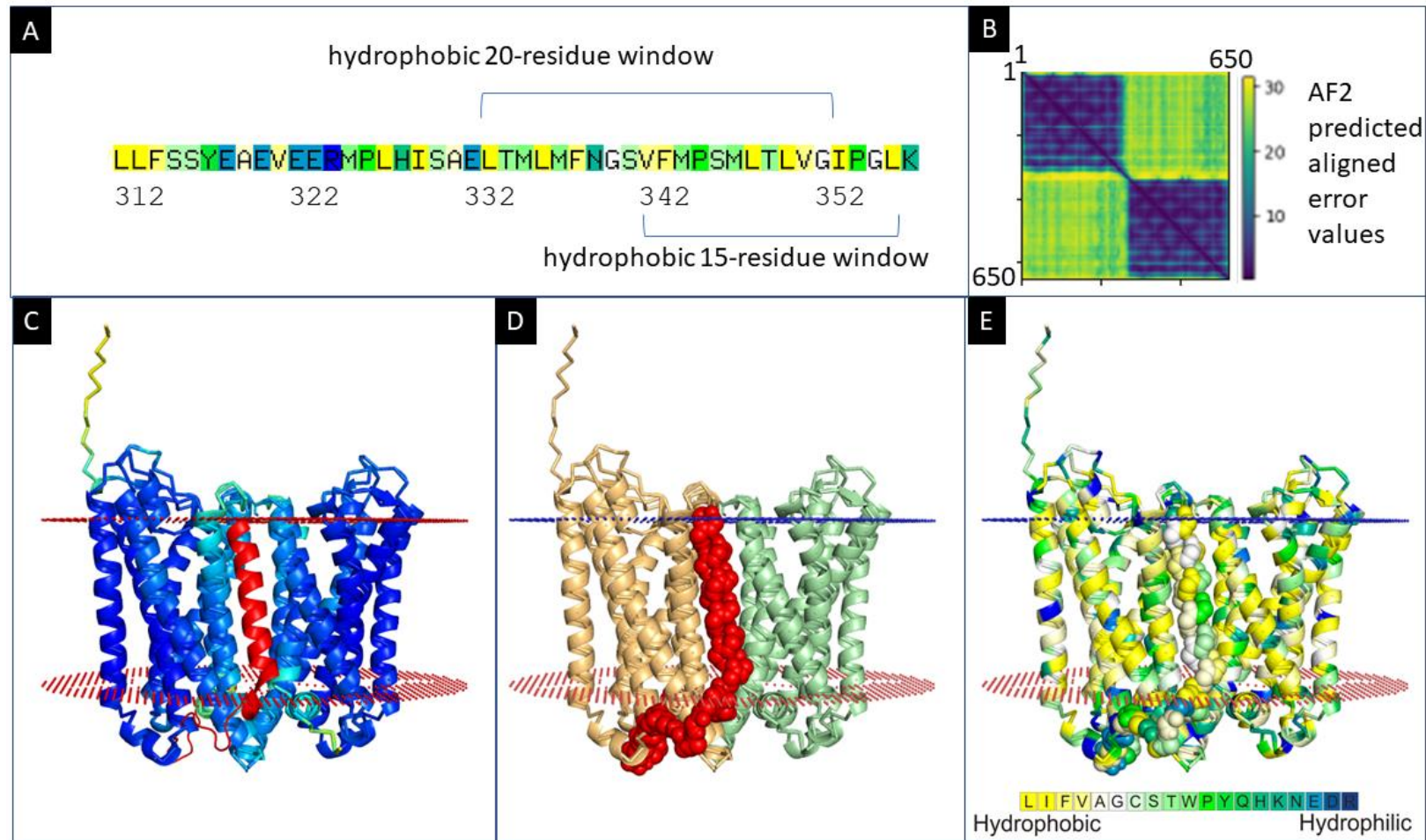
vacuum		
$\text{N-H} + \text{O=C}$	$\xrightarrow{\Delta E_1 = -6.6}$	$\text{N-H} \cdots \text{O=C}$
$\Delta G_2^{\text{elec}} = -10.3$		$\Delta G_3^{\text{elec}} = -4.8$
$\Delta G_2^{\text{bp}} = 0.3$		$\Delta G_3^{\text{bp}} = 0.1$
$\Delta G_2 = -10.0$		$\Delta G_3 = -4.7$
water		
$\text{N-H} + \text{O=C}$	$\xrightarrow{\Delta G_4 = -1.3}$	$\text{N-H} \cdots \text{O=C}$
$\Delta G_5^{\text{elec}} = 9.3$		$\Delta G_6^{\text{elec}} = 4.2$
$\Delta G_5^{\text{bp}} = -1.5$		$\Delta G_6^{\text{bp}} = -0.4$
$\Delta G_5 = 7.8$		$\Delta G_6 = 3.8$
organic solvent		
$\text{N-H} + \text{O=C}$	$\xrightarrow{\Delta G_7 = -5.3}$	$\text{N-H} \cdots \text{O=C}$



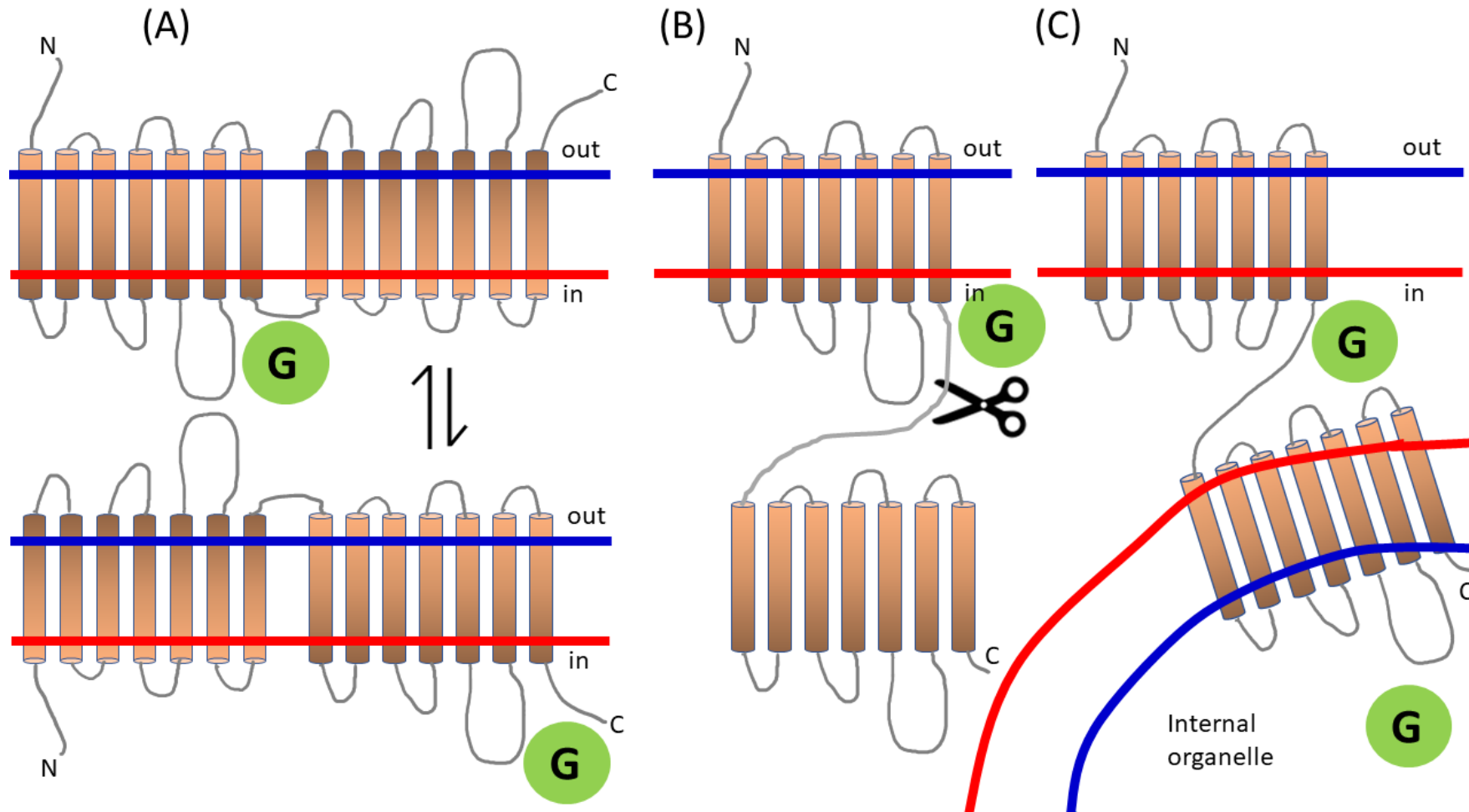
The linkers connecting the GPCR domains should be hydrophobic TM helices

But most linkers are not sufficiently long and hydrophobic...

Predicted structure of two-GPCR-domain A0A3L7HXX2 (from rodent)



Speculative scenarios for a two-GPCR protein with a linker that is too polar to cross the membrane



Summary and outlook

- Many putative proteins with multi-GPCR domains
- From fish to mammals
- Up to monkeys
- Annotated mostly as olfactory receptors; few chemokine receptors
- None of them prevail, so probably not indispensable
- What is their unique contribution?
- What is their membrane topology?
- Mechanisms?
- Multi-barrel and multi-GPCR domains... What other multi-domain membrane proteins are there?
- Fragmented GPCR domains?

Acknowledgments

The Ben-Tal lab: <https://www.bentalab.com/>



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Ron Solan

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Rachel Kolodny

Funding:



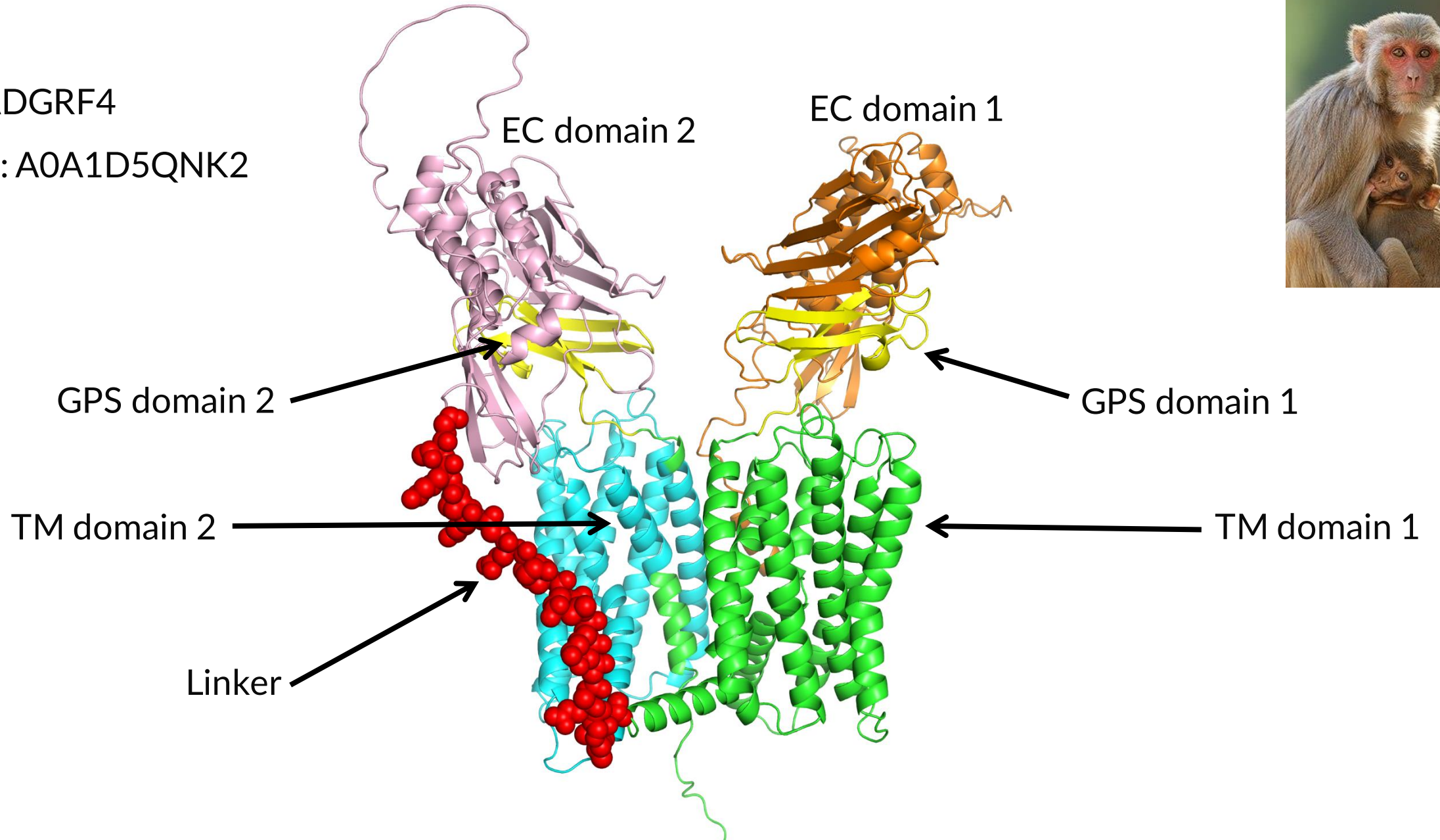
Volkswagen **Stiftung**



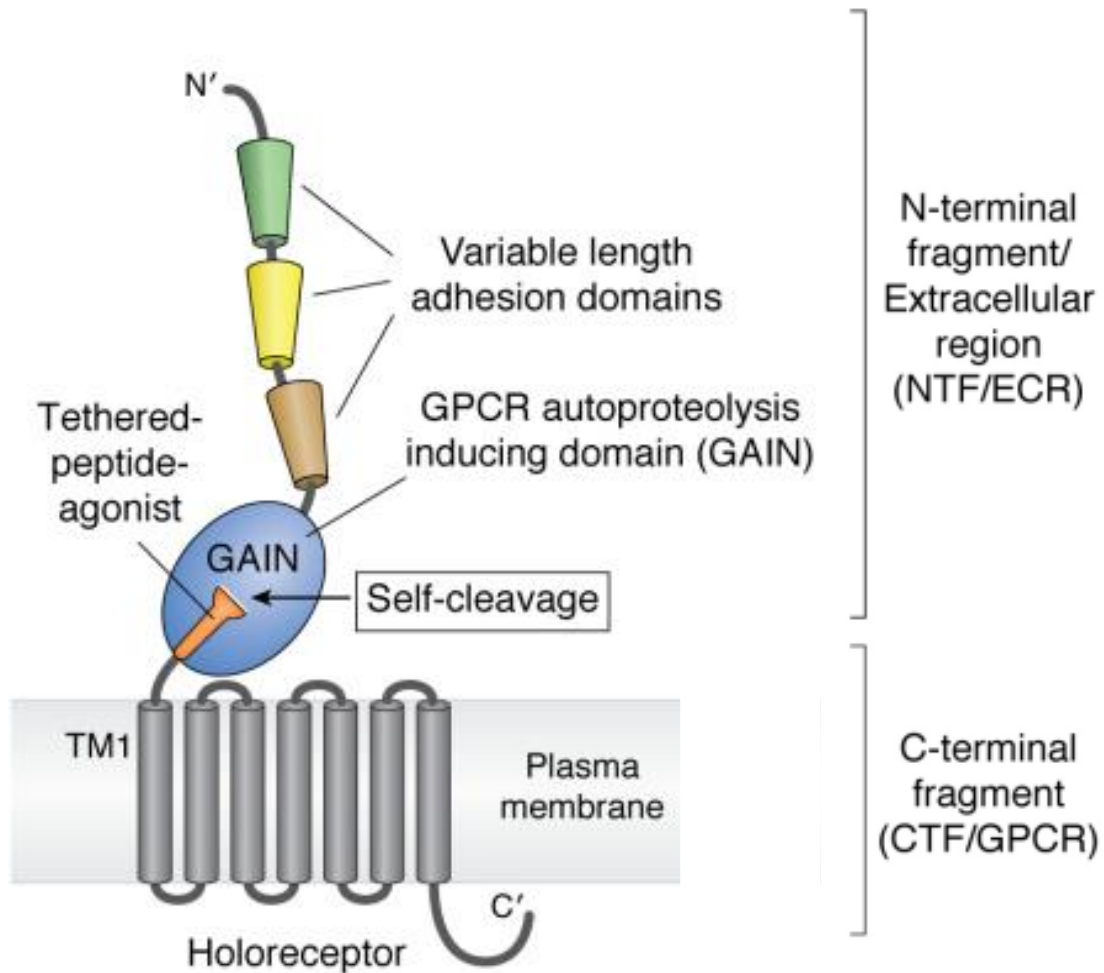
Adhesion GPCR F4 from Rhesus Macaque

Gene: ADGRF4

UniProt: A0A1D5QNK2



Adhesion GPCR F4 from Rhesus Macaque

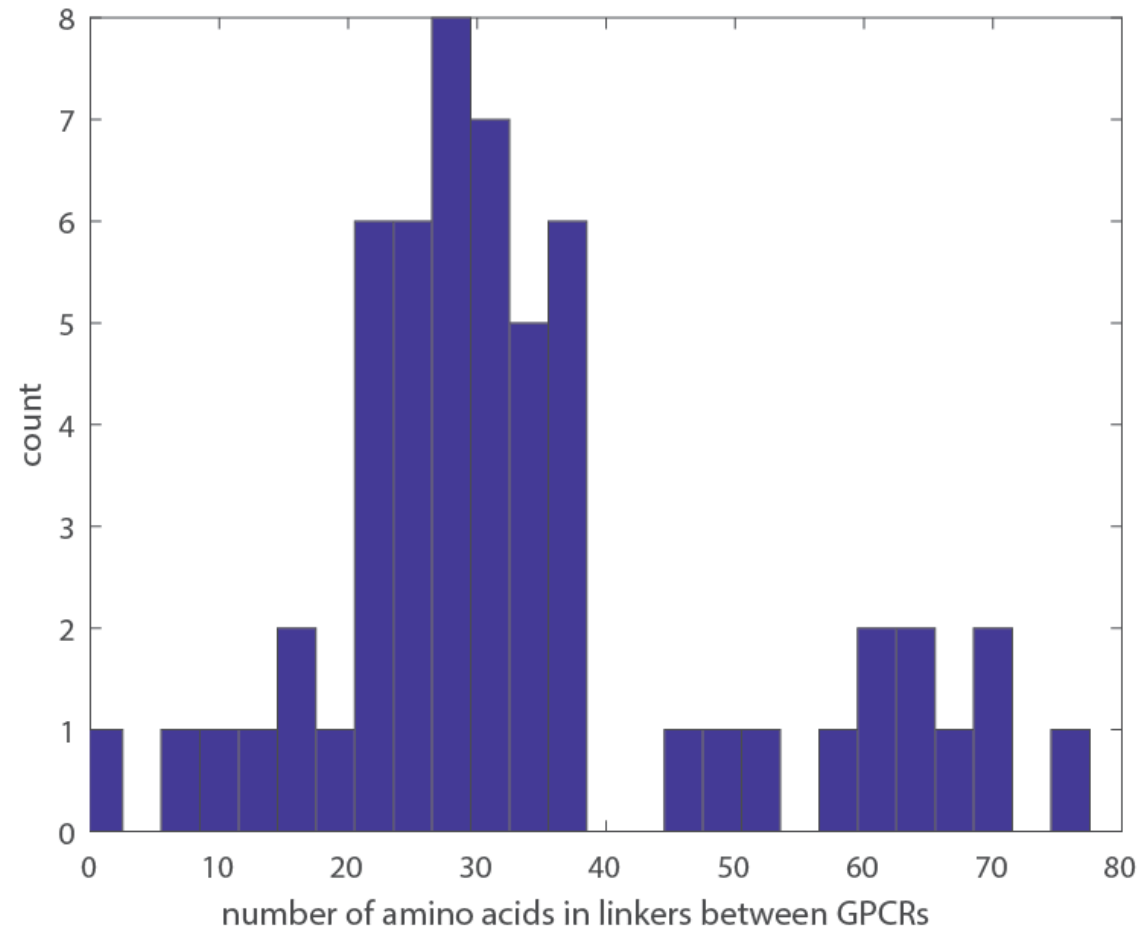


HLS/T motif

Human AGRF3 *	C	Q	-	-	-	H	L	T	A
Human AGRF4	C	N	Y	T	S	V	V	M	S
A0A1D5QNK2_1	C	R	P	S	K	L	F	T	S
A0A1D5QNK2_2	C	N	Y	T	S	T	V	M	S
F7ELW9_1	C	R	P	S	K	L	F	T	S
F7ELW9_2	C	N	Y	T	S	T	V	M	S

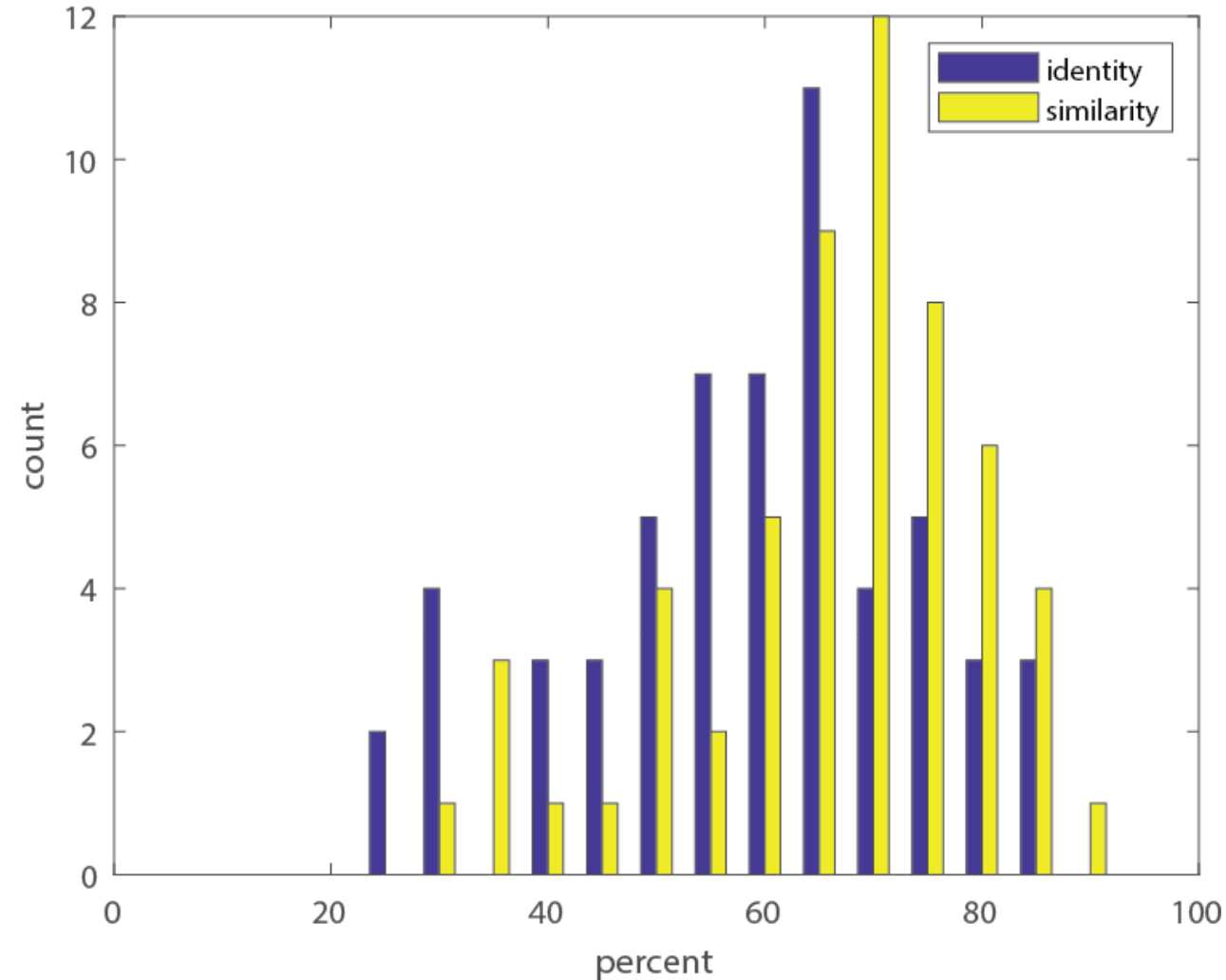
* Auto-proteolysed

Distribution of the number of residues in the linker regions between the different GPCR domains in the same chain



Isildayancan, Kessel, Solan, Kolody, Ben-Tal. *BioRxiv*.

Distribution of sequence identify/similarity among the different GPCR domains within the same chain



Isildayancan, Kessel, Solan, Kolody, Ben-Tal. *BioRxiv*.